

A Statistical Analysis of Winning in College Baseball

Wyatt Thompson

Purpose

The goal of this report is to provide a comprehensive analysis on what factors contribute to successful teams in college baseball. We initially will examine the 2026 Southeastern Conference, as time and time again the SEC has proved itself to be the pinnacle of college baseball. Further analysis will be done later.

Defining a Successful Team

- We will define “successful teams” as teams with high win percentages (e.g., a team with a 0.65 win percentage will be considered more successful than one with a 0.54 win percentage) since there is a wide discrepancy between teams in raw number of games played
- The following analysis can be done using win percentage as the response variable; however, we will use a different unit
 - We will calculate the average number of games played by team in the SEC, and transform the win percentage into expected wins if every team played the same number of games; this will make the end result easier to interpret

The Data

Data Definitions

- BA – batting average – hits per at-bat
- SLG. – slugging – bases per at-bat (extra weight to doubles, triples, HRs)
- OB. – on-base percentage – percentage of at-bats where batters reached a base
- BB – walks – the number of times batters were walked
- K – strikeouts – the number of times batters struck out
- X2B – doubles – the number of times batters hit a double
- X3B – triples – the number of times batters hit a triple
- HR – homeruns – the number of times batters hit a homerun
- ERA – earned run average – the average number of earned runs the pitching staff gave up per 9 innings
- WHIP – walks + hits per inning – the average number of walks and hits the pitching staff gave up per inning

Standings

There are 16 teams in the SEC. We will gather some basic data for each team first – the team’s name, the team’s number of wins and losses, its number of games played, and its win percentage. Then, we will calculate the average number of games each team played and multiply that number by each team’s win percentage.

The average number of games played by SEC teams was 59.625.

Table 1: SEC Standings

Team	Wins	Losses	Games	WinPct	MeanWins
Georgia	51	12	63	0.810	48.29625
Texas	45	13	58	0.776	46.26900
Texas A&M	41	16	57	0.719	42.87038
Mississippi State	43	17	60	0.717	42.75113
Alabama	42	19	61	0.689	41.08163
Florida	41	21	62	0.661	39.41213
Ole Miss	41	21	62	0.661	39.41213
Auburn	42	22	64	0.656	39.11400
Arkansas	41	22	63	0.651	38.81587
Tennessee	38	22	60	0.633	37.74262
Oklahoma	38	22	60	0.633	37.74262
Kentucky	33	23	56	0.589	35.11912
Vanderbilt	33	25	58	0.569	33.92663
LSU	30	28	58	0.517	30.82613
Missouri	24	31	55	0.436	25.99650
South Carolina	22	35	57	0.386	23.01525

Batting Data

The next step is to grab batting data. We will likely start out with too many variables, but that is okay. We will see which ones are predictive of wins and which are just noise.

Table 2: Batting Stats

Stat	Name
AVG	Batting average
SLG	Slugging
OB	On-base percentage
BB	Walks
K	Strikeouts
X2B	Doubles
X3B	Triples
HR	Homeruns

Table 3: Batting Data

Team	AVG	SLG.	OB.	BB	K	X2B	X3B	HR
Georgia	0.326	0.628	0.439	334	497	122	5	174
Mississippi State	0.317	0.558	0.415	292	495	133	6	117
Texas A&M	0.301	0.564	0.420	322	473	116	10	121
Texas	0.298	0.521	0.420	373	487	109	7	103
Auburn	0.295	0.477	0.390	274	490	108	5	90
Vanderbilt	0.290	0.520	0.399	307	550	110	10	108
Oklahoma	0.289	0.484	0.392	302	513	100	17	83
Kentucky	0.284	0.458	0.398	230	457	110	14	62
LSU	0.282	0.482	0.397	306	468	101	7	91
Florida	0.279	0.495	0.384	287	496	110	11	106
Tennessee	0.277	0.514	0.367	239	483	127	7	115
Arkansas	0.275	0.484	0.379	316	605	114	6	105
Ole Miss	0.264	0.468	0.385	331	657	89	1	107
Missouri	0.261	0.407	0.371	223	455	78	12	52
Alabama	0.258	0.440	0.377	290	556	101	10	80
South Carolina	0.235	0.407	0.325	194	560	103	8	68

Pitching Data

Next we will gather pitching data. A lot of the variables are related to each other, so we will have fewer variables here. We will also not include walks from a pitching standpoint since that is included in WHIP.

Table 4: Pitching Stats

Stat	Name
ERA	Earned run average
WHIP	Walks plus hits per inning
SO	Strikeouts

Table 5: Pitching Stats

Team	ERA	WHIP	SO
Auburn	3.69	1.18	645
Texas	4.08	1.22	658
Alabama	4.11	1.32	532
Mississippi State	4.26	1.32	663
Ole Miss	4.37	1.32	688
Florida	4.61	1.32	636
Tennessee	4.72	1.27	614
Arkansas	4.74	1.37	631
South Carolina	4.81	1.49	489
Georgia	4.92	1.43	653
Oklahoma	5.15	1.39	596
Vanderbilt	5.23	1.43	563
Texas A&M	5.24	1.41	497
Kentucky	5.51	1.51	508
LSU	5.72	1.50	669
Missouri	5.75	1.55	520

Analysis

Our method of analysis will be linear regression. We will use all the previously mentioned variables as predictors and mean wins as the response variable.

```
##
## Call:
## lm(formula = MeanWins ~ AVG + SLG. + OB. + BB + K + X2B + X3B +
##      HR + ERA + WHIP + SO, data = df)
##
## Residuals:
##      1      2      3      4      5      6      7      8      9     10
## 0.4458 0.2981 0.9123 0.4191 -0.2674 -1.1823 0.3766 -0.8309 1.7296 0.6109
##     11     12     13     14     15     16
## 1.1839 0.1146 -3.6633 -0.9677 0.9902 -0.1696
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.438e+00  3.162e+01  -0.077   0.9422
## AVG          -1.895e+02  1.963e+02  -0.965   0.3892
## SLG.          -9.652e+01  1.505e+02  -0.641   0.5562
## OB.           3.683e+02  1.271e+02   2.897   0.0443 *
## BB            -2.516e-02  4.805e-02  -0.524   0.6282
## K              1.056e-02  2.219e-02   0.476   0.6589
## X2B            1.477e-01  1.257e-01   1.174   0.3054
## X3B            5.610e-01  2.923e-01   1.919   0.1274
## HR             2.176e-01  1.816e-01   1.198   0.2971
## ERA            -2.125e+00  3.454e+00  -0.615   0.5716
## WHIP           -2.397e+01  1.904e+01  -1.259   0.2766
## SO             2.686e-04  2.336e-02   0.012   0.9914
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.436 on 4 degrees of freedom
## Multiple R-squared:  0.965, Adjusted R-squared:  0.8689
## F-statistic: 10.04 on 11 and 4 DF, p-value: 0.01965
```

Interpretations

Quick warning: The following results are going to seem intuitively wrong. That's okay. We will fix that later.

None of the variables in the model are individually statistically significant outside of OBP. However, the p-value of 0.01965 for our F-statistic does indicate that the model as a whole *is* significant; the adjusted R^2 of 0.869 also tells us that the model does a good job at explaining the variance in wins between teams.

Now comes the fun part. Let's look at the variables' coefficients – or in other words, how much each variable impacts winning. Attached is a table showing these effects.

Table 6: Results

Stat	Value	Interpretation
AVG	-189.5000	0.10 increase in batting average correlated with ~19 fewer wins
SLG.	-96.5000	0.10 increase in slugging correlated with ~10 fewer wins
OB.	368.3000	0.10 increase in on-base percentage correlated with ~37 more wins
BB	-0.0250	100 BB's is worth about -3 wins
K	0.0110	100 K's (batter) is worth about 1 win
X2B	0.1480	10 doubles are worth about 1 win
X3B	0.5610	10 triples are worth about 6 wins
HR	0.2180	10 homeruns are worth about 2 wins
ERA	-2.1250	An ERA decrease of 1 is worth about 2 wins
WHIP	-0.2400	A WHIP improvement of 1 is worth about 0.2 wins
SO	0.0003	Very little impact from SO's on wins

Now, obviously, some of this does not make sense. Why would increasing batting average and slugging decrease wins? This is where multicollinearity comes into play. This is when variables are intrinsically related to each other in a way that is detrimental to the integrity of the model. For example, slugging is directly calculated from doubles, triples, and homeruns. Additionally, on-base percentage is partially dependent on batting average.

We can measure how much multicollinearity affects the variance of our coefficients using the Variance Inflation Factor (VIF). A VIF under 5 is solid, a VIF under 10 is cautioned but acceptable, and a VIF over 10 means there is too much correlation between variables.

Table 7: VIF

	x
AVG	49.838847
SLG.	188.516778
OB.	28.515203
BB	12.853477
K	3.954223
X2B	7.305741
X3B	3.281628
HR	68.010681
ERA	11.089312
WHIP	10.404696
SO	6.511546

This confirms we have a multicollinearity problem. This is especially evident with slugging and homeruns. Because slugging is calculated using extra base hits, we will remove doubles, triples, and homeruns from the model in favor of just slugging. Furthermore, we will remove batting average and walks in favor of on-base percentage since they are also directly related.

```
##
## Call:
## lm(formula = MeanWins ~ SLG. + OB. + K + ERA + WHIP + SO, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.9827 -1.0734 -0.1368  1.6583  2.9767
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.40059    22.01703   0.064  0.9507
## SLG.          29.07275    22.05165   1.318  0.2199
## OB.          156.72252    49.89615   3.141  0.0119 *
## K              0.02023     0.01553   1.302  0.2251
## ERA          -1.51315     2.98821  -0.506  0.6248
## WHIP        -24.50141    17.64167  -1.389  0.1983
## SO           -0.01477     0.01386  -1.066  0.3143
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.712 on 9 degrees of freedom
## Multiple R-squared:  0.9025, Adjusted R-squared:  0.8376
## F-statistic: 13.89 on 6 and 9 DF,  p-value: 0.0004263
```

This is our new model. We need to ensure that we eliminated the multicollinearity.

Table 8: VIF

	x
SLG.	3.267364
OB.	3.544170
K	1.563588
ERA	6.698315
WHIP	7.208582
SO	1.851172

This model is much better. There are some moderately high values but none are in the “danger zone”. Our adjusted R^2 did not decline by much, and our p-value is magnitudes times more significant now. Now we should look again at coefficients.

Table 9: Results

Stat	Value	Interpretation
SLG.	29.073	A 0.10 increase in slugging correlates with 2.9 more wins
OB.	156.722	A 0.10 increase in on-base percentage correlates with 15.7 more wins
K	0.020	100 strikeouts (batter) is worth 2 wins
ERA	-1.513	A 1.00 improvement in ERA is worth 1.5 wins
WHIP	-24.501	A 0.25 improvement in WHIP is worth 6.125 wins
SO	-0.015	100 strikeouts (pitcher) is worth -1.5 wins

These results intuitively make a lot of sense. Better slugging and on-base percentage *should* lead to more wins, as should lower ERA and WHIP numbers. The one aspect that is questionable is strikeouts. Our coefficients indicate that when a team's batters have more strikeouts, that team wins more; they also indicate that when a team's pitchers throw more strikeouts, that team wins less. This makes slightly more sense when you consider that teams that swing for power will generally strikeout more, but those strikeouts get outweighed by the extra base hits that come with it. On the flip side, pitching strikeouts likely negatively correlates with wins because it generally requires more pitches to be thrown, and therefore more opportunities for runs to be scored. That is the theory, anyways.

Next Steps – The ACC

The ACC has arguably been the next best conference in college baseball over recent years. Teams such as North Carolina, Wake Forest, and Georgia Tech have had great seasons recently. We will first perform the same analysis that we did on the SEC on the ACC in order to find potential differences between conferences; we then will combine all the data and model that to have a larger sample size.

The average number of games played by ACC teams was 57.9375.

Note: we are not including North Carolina's tied game in the standings

Table 10: ACC Standings

Team	Wins	Losses	Games	WinPct	MeanWins
Georgia Tech	50	11	61	0.820	47.50875
North Carolina	50	12	62	0.806	46.69763
Florida State	40	19	59	0.678	39.28162
Miami	39	20	59	0.661	38.29669
Wake Forest	39	21	60	0.650	37.65937
Boston College	40	24	64	0.625	36.21094
Virginia	37	23	60	0.617	35.74744
Notre Dame	31	22	53	0.585	33.89344
Pitt	33	24	57	0.579	33.54581
NC State	32	24	56	0.571	33.08231
Clemson	31	26	57	0.544	31.51800
Virginia Tech	30	26	56	0.536	31.05450
California	29	26	55	0.527	30.53306
Louisville	30	27	57	0.526	30.47512
Stanford	28	26	54	0.519	30.06956
Duke	26	31	57	0.456	26.41950

Table 11: Batting Data

Team	SLG.	OB.	K
Georgia Tech	0.624	0.467	442
Notre Dame	0.512	0.412	462
NC State	0.499	0.414	374
Louisville	0.508	0.417	470
Pitt	0.534	0.435	454
Miami	0.493	0.402	430
Wake Forest	0.508	0.407	467
North Carolina	0.483	0.413	458
Stanford	0.493	0.386	445
Clemson	0.449	0.387	416
Florida State	0.468	0.382	542
Virginia Tech	0.474	0.389	492
Virginia	0.488	0.397	560
Boston College	0.424	0.387	437
Duke	0.463	0.373	506
California	0.446	0.346	466

Table 12: Pitching Data

Team	ERA	WHIP	SO
North Carolina	3.87	1.359	590
Florida State	3.97	1.265	630
California	4.37	1.429	471
Clemson	4.56	1.321	558
Wake Forest	4.70	1.411	677
Georgia Tech	4.75	1.406	550
Boston College	4.94	1.505	492
Miami	5.03	1.466	576
Virginia	5.45	1.513	593
NC State	5.93	1.493	552
Stanford	5.98	1.705	473
Notre Dame	6.22	1.582	469
Pitt	6.35	1.671	442
Louisville	6.75	1.807	493
Duke	6.81	1.668	455
Virginia Tech	6.95	1.695	519

Now that we have all of our data, let's model wins.

```
##
## Call:
## lm(formula = MeanWins ~ SLG. + OB. + K + ERA + WHIP + SO, data = df_acc)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.8443 -1.1449  0.0608  1.0632  3.3212
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -19.965005   19.147320  -1.043   0.3243
## SLG.         -13.027177   32.158081  -0.405   0.6949
## OB.          166.158621   56.384060   2.947   0.0163 *
## K              0.022568    0.016803   1.343   0.2121
## ERA          -4.622110    1.645652  -2.809   0.0204 *
## WHIP          4.920014   11.833841   0.416   0.6873
## SO            0.003696    0.014643   0.252   0.8064
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.563 on 9 degrees of freedom
## Multiple R-squared:  0.8828, Adjusted R-squared:  0.8047
## F-statistic: 11.3 on 6 and 9 DF,  p-value: 0.0009434
```

These results are quite interesting. We see a similar effect from on-base percentage – ~17 more wins for a 0.10 increase – but we actually see a negative correlation between slugging and wins in the ACC. My working theory is that, since we can see ERA is even more important here, a lot of the teams with better hitting had worse pitching. Regardless, once we combine conferences into one big sample I expect slugging to be worth 2-3 wins for a 0.10 increase.

```
##
## Call:
## lm(formula = MeanWins ~ SLG. + OB. + K + ERA + WHIP + SO, data = master_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.2548 -1.5675 -0.1564  1.8464  4.0019
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -8.254367   13.728242  -0.601 0.553074
## SLG.          22.989727   16.958721   1.356 0.187338
## OB.         137.788692   34.046125   4.047 0.000439 ***
## K              0.017096    0.009895   1.728 0.096381 .
## ERA          -3.445534    1.340555  -2.570 0.016507 *
## WHIP          -7.176851    8.863504  -0.810 0.425748
## SO           -0.002871    0.009621  -0.298 0.767888
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.59 on 25 degrees of freedom
## Multiple R-squared:  0.8642, Adjusted R-squared:  0.8316
## F-statistic: 26.52 on 6 and 25 DF,  p-value: 1.084e-09
```

After combining the ACC and SEC, we see that a 0.10 increase in slugging is now worth about 2.3 wins. On-base percentage is still a heavy-hitter with a 0.10 increase being worth about 14 wins. A 1.00 improvement in ERA is worth 3.4 wins and a 0.25 improvement in WHIP is worth nearly 2 wins. As I expected, increasing the sample size steadied these numbers. I am going to add in Big Ten and Big 12 stats next, but for brevity I will skip straight to the modeling sections. I will leave individual conference modeling here in case there is any interest, but I won't comment on it until I get to the big sample model.

Big 12 Model

```
##
## Call:
## lm(formula = MeanWins ~ SLG. + OB. + K + ERA + WHIP + SO, data = df_12)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-2.9136	-1.4613	-0.3628	1.2978	5.8943

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-29.678562	29.068503	-1.021	0.341
SLG.	31.406403	50.234366	0.625	0.552
OB.	154.294237	81.992953	1.882	0.102
K	0.017053	0.012373	1.378	0.211
ERA	-5.772620	3.636374	-1.587	0.156
WHIP	7.959700	22.406753	0.355	0.733
SO	-0.002528	0.018900	-0.134	0.897

```
##
## Residual standard error: 3.021 on 7 degrees of freedom
## Multiple R-squared: 0.87, Adjusted R-squared: 0.7586
## F-statistic: 7.81 on 6 and 7 DF, p-value: 0.007919
```

SEC, ACC, Big 12 Model

```
##
## Call:
## lm(formula = MeanWins ~ SLG. + OB. + K + ERA + WHIP + SO, data = master_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.0341 -1.8276 -0.0274  1.7691  4.9012
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.435e+01  1.066e+01  -1.347  0.18578
## SLG.         2.501e+01  1.426e+01   1.754  0.08732 .
## OB.          1.400e+02  2.801e+01   4.999 1.26e-05 ***
## K             1.658e-02  6.836e-03   2.425  0.02004 *
## ERA          -3.856e+00  1.160e+00  -3.323  0.00194 **
## WHIP         -3.903e+00  7.430e+00  -0.525  0.60234
## SO            1.443e-04  6.951e-03   0.021  0.98355
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.487 on 39 degrees of freedom
## Multiple R-squared:  0.8731, Adjusted R-squared:  0.8535
## F-statistic: 44.71 on 6 and 39 DF,  p-value: 5.635e-16
```

Big Ten Model

```
##
## Call:
## lm(formula = MeanWins ~ SLG. + OB. + K + ERA + WHIP + SO, data = standings_10)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.4728 -1.1758  0.1521  1.4139  5.4858
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -28.979687  28.737033  -1.008  0.3370
## SLG.         15.838382  40.746116   0.389  0.7056
## OB.          162.484004  65.671929   2.474  0.0329 *
## K             0.024542   0.026448   0.928  0.3753
## ERA          -7.802458   3.149808  -2.477  0.0327 *
## WHIP         17.421300  19.153240   0.910  0.3844
## SO           -0.008195   0.018493  -0.443  0.6671
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.444 on 10 degrees of freedom
## Multiple R-squared:  0.8867, Adjusted R-squared:  0.8188
## F-statistic: 13.05 on 6 and 10 DF,  p-value: 0.0003213
```

SEC, ACC, Big 12, Big 10 Model

```
##
## Call:
## lm(formula = MeanWins ~ SLG. + OB. + K + ERA + SO, data = master_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.7966 -1.8760 -0.1159  1.8260  5.1346
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.712e+01  7.440e+00  -2.301   0.0251 *
## SLG.         2.503e+01  1.282e+01   1.953   0.0558 .
## OB.          1.416e+02  2.422e+01   5.846 2.56e-07 ***
## K             1.686e-02  6.484e-03   2.600   0.0118 *
## ERA          -4.573e+00  4.338e-01 -10.543 5.22e-15 ***
## SO            6.073e-05  5.835e-03   0.010   0.9917
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.57 on 57 degrees of freedom
## Multiple R-squared:  0.8793, Adjusted R-squared:  0.8687
## F-statistic: 83.05 on 5 and 57 DF,  p-value: < 2.2e-16
```

Note: I removed WHIP for the full model because, after testing for multicollinearity, ERA and WHIP were both above a VIF of 10 in this final model. After removing WHIP, all the VIF numbers were below 3.5—an excellent sign

Final Conclusions

Once we reached a sample size of 63 teams, our numbers stabilized. Our model's adjusted R^2 of 0.8687 is excellent, and the model's overall p-value is extremely significant. Most individual p-value for our variables got smaller, which is a sign of significance. On-base percentage, batter strikeouts, and ERA are all significant at a 5% level, and slugging is significant at a 10% level. Pitching strikeouts by far has the least impact on winning. Below is one final interpretation table.

Table 13: Win Values in Baseball

Stat	Value	Interpretation
SLG.	25.03000	0.10 increase in slugging is worth 2.5 wins
OB.	141.60000	0.10 increase in on-base percentage is worth 14 wins
K	0.01686	100 batting strikeouts is worth 1.7 wins
ERA	-4.57300	Improvement of 1.00 in ERA is worth 4.6 wins
SO	0.00006	Pitching strikeouts have almost no effect on wins

For those curious, diagnostic plots confirmed all of our linear regression assumptions were satisfied, which further validates this analysis.

Actionable Insights

What can college teams learn from this analysis? College baseball teams tend to struggle with funding, so it is important to efficiently build teams for winning. You cannot afford many misses. Teams looking to win, at least for these four major conferences, should have a primary focus on getting on base and reducing earned runs. Slugging should come third to to these stats. Another insight is that strikeouts, **on their own**, do not matter as much as we may think—batters striking out does not hinder winning and pitchers getting strikeouts has almost no impact. What is more important is likely pitching efficiency. Groundouts will get the job done.

Source

Data was collected from the official websites of the SEC, ACC, Big Ten, and Big 12.